

```

swaraj@swaraj-VirtualBox:~$ mongo
MongoDB shell version v4.4.29
connecting to: mongoddb://127.0.0.1:27017/?
compressors=disabled&gssapiServiceName=mongoddb
Implicit session: session { "id" : UUID("0c7d61d3-c800-441c-8478-
d46cc31fff33") }
MongoDB server version: 4.4.29
---
The server generated these startup warnings when booting:
    2025-09-15T11:58:03.988+05:30: Using the XFS filesystem is strongly
recommended with the WiredTiger storage engine. See
http://dochub.mongodb.org/core/prodnotes-filesystem
    2025-09-15T11:58:05.184+05:30: Access control is not enabled for the
database. Read and write access to data and configuration is unrestricted
---
```

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> use studentDB
switched to db studentDB
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```

> db.createCollection("student")
{ "ok" : 1 }
```

```

> db.student.insertMany([
...   { name: "Amit", age: 20, gender: "male", class: "TE A", CGPA: 7.2, city:
"Pune", course: "CS", graduated: false },
...   { name: "Priya", age: 19, gender: "female", class: "SE B", CGPA: 8.1,
city: "Bangalore", course: "IT", graduated: true },
...   { name: "Rahul", age: 21, gender: "male", class: "TE B", CGPA: 6.4, city:
"Mumbai", course: "CS", graduated: false },
...   { name: "Sayali", age: 22, gender: "female", class: "BE A", CGPA: 7.8,
city: "Nagpur", course: "BCA", graduated: true },
...   { name: "Vikram", age: 18, gender: "male", class: "FE A", CGPA: 5.9, city:
"Delhi", course: "Mechanical", graduated: false },
...   { name: "Anjali", age: 20, gender: "female", class: "TE A", CGPA: 9.0,
city: "Bangalore", course: "CS", graduated: true },
...   { name: "Karan", age: 19, gender: "male", class: "SE A", CGPA: 7.5, city:
"Hyderabad", course: "IT", graduated: true },
...   { name: "Neha", age: 18, gender: "female", class: "FE B", CGPA: 6.8, city:
"Chennai", course: "CS", graduated: false },
...   { name: "Suresh", age: 21, gender: "male", class: "BE B", CGPA: 8.5, city:
"Bangalore", course: "Civil", graduated: true },
...   { name: "Pooja", age: 20, gender: "female", class: "TE B", CGPA: 7.0,
city: "Pune", course: "CS", graduated: false },
...   { name: "Rohit", age: 23, gender: "male", class: "BE A", CGPA: 6.0, city:
"Nagpur", course: "IT", graduated: false }
... ]);
{
  "acknowledged" : true,
  "insertedIds" : [
    ObjectId("68c7b53642e9879892378990"),
    ObjectId("68c7b53642e9879892378991"),
    ObjectId("68c7b53642e9879892378992"),
    ObjectId("68c7b53642e9879892378993"),
    ObjectId("68c7b53642e9879892378994"),
    ObjectId("68c7b53642e9879892378995"),
    ObjectId("68c7b53642e9879892378996"),
    ObjectId("68c7b53642e9879892378997"),
    ObjectId("68c7b53642e9879892378998"),
    ObjectId("68c7b53642e9879892378999"),
    ObjectId("68c7b53642e987989237899a")
  ]
}

> db.student.find().pretty()
```

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{
  "_id" : ObjectId("68c7b53642e9879892378990"),
  "name" : "Amit",
  "age" : 20,
  "gender" : "male",
  "class" : "TE A",
  "CGPA" : 7.2,
  "city" : "Pune",
  "course" : "CS",
  "graduated" : false
}
{
  "_id" : ObjectId("68c7b53642e9879892378991"),
  "name" : "Priya",
  "age" : 19,
  "gender" : "female",
  "class" : "SE B",
  "CGPA" : 8.1,
  "city" : "Bangalore",
  "course" : "IT",
  "graduated" : true
}
{
  "_id" : ObjectId("68c7b53642e9879892378992"),
  "name" : "Rahul",
  "age" : 21,
  "gender" : "male",
  "class" : "TE B",
  "CGPA" : 6.4,
  "city" : "Mumbai",
  "course" : "CS",
  "graduated" : false
}
{
  "_id" : ObjectId("68c7b53642e9879892378993"),
  "name" : "Sayali",
  "age" : 22,
  "gender" : "female",
  "class" : "BE A",
  "CGPA" : 7.8,
  "city" : "Nagpur",
  "course" : "BCA",
  "graduated" : true
}
{
  "_id" : ObjectId("68c7b53642e9879892378994"),
  "name" : "Vikram",
  "age" : 18,
  "gender" : "male",
  "class" : "FE A",
  "CGPA" : 5.9,
  "city" : "Delhi",
  "course" : "Mechanical",
  "graduated" : false
}
{
  "_id" : ObjectId("68c7b53642e9879892378995"),
  "name" : "Anjali",
  "age" : 20,
  "gender" : "female",
  "class" : "TE A",
  "CGPA" : 9,
  "city" : "Bangalore",
  "course" : "CS",

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    "graduated" : true
  }
  {
    "_id" : ObjectId("68c7b53642e9879892378996"),
    "name" : "Karan",
    "age" : 19,
    "gender" : "male",
    "class" : "SE A",
    "CGPA" : 7.5,
    "city" : "Hyderabad",
    "course" : "IT",
    "graduated" : true
  }
  {
    "_id" : ObjectId("68c7b53642e9879892378997"),
    "name" : "Neha",
    "age" : 18,
    "gender" : "female",
    "class" : "FE B",
    "CGPA" : 6.8,
    "city" : "Chennai",
    "course" : "CS",
    "graduated" : false
  }
  {
    "_id" : ObjectId("68c7b53642e9879892378998"),
    "name" : "Suresh",
    "age" : 21,
    "gender" : "male",
    "class" : "BE B",
    "CGPA" : 8.5,
    "city" : "Bangalore",
    "course" : "Civil",
    "graduated" : true
  }
  {
    "_id" : ObjectId("68c7b53642e9879892378999"),
    "name" : "Pooja",
    "age" : 20,
    "gender" : "female",
    "class" : "TE B",
    "CGPA" : 7,
    "city" : "Pune",
    "course" : "CS",
    "graduated" : false
  }
  {
    "_id" : ObjectId("68c7b53642e987989237899a"),
    "name" : "Rohit",
    "age" : 23,
    "gender" : "male",
    "class" : "BE A",
    "CGPA" : 6,
    "city" : "Nagpur",
    "course" : "IT",
    "graduated" : false
  }
}

> db.student.find({
...   $or: [
...     { $and: [{ gender: "male" }, { class: "TE A" }] },
...     { CGPA: { $gte: 7 } }
...   ]
... });

```

```

{ "_id" : ObjectId("68c7b53642e9879892378990"), "name" : "Amit", "age" : 20,
"gender" : "male", "class" : "TE A", "CGPA" : 7.2, "city" : "Pune", "course" :
"CS", "graduated" : false }
{ "_id" : ObjectId("68c7b53642e9879892378991"), "name" : "Priya", "age" : 19,
"gender" : "female", "class" : "SE B", "CGPA" : 8.1, "city" : "Bangalore",
"course" : "IT", "graduated" : true }
{ "_id" : ObjectId("68c7b53642e9879892378993"), "name" : "Sayali", "age" : 22,
"gender" : "female", "class" : "BE A", "CGPA" : 7.8, "city" : "Nagpur", "course" :
"BCA", "graduated" : true }
{ "_id" : ObjectId("68c7b53642e9879892378995"), "name" : "Anjali", "age" : 20,
"gender" : "female", "class" : "TE A", "CGPA" : 9, "city" : "Bangalore",
"course" : "CS", "graduated" : true }
{ "_id" : ObjectId("68c7b53642e9879892378996"), "name" : "Karan", "age" : 19,
"gender" : "male", "class" : "SE A", "CGPA" : 7.5, "city" : "Hyderabad",
"course" : "IT", "graduated" : true }
{ "_id" : ObjectId("68c7b53642e9879892378998"), "name" : "Suresh", "age" : 21,
"gender" : "male", "class" : "BE B", "CGPA" : 8.5, "city" : "Bangalore",
"course" : "Civil", "graduated" : true }
{ "_id" : ObjectId("68c7b53642e9879892378999"), "name" : "Pooja", "age" : 20,
"gender" : "female", "class" : "TE B", "CGPA" : 7, "city" : "Pune", "course" :
"CS", "graduated" : false }

> db.student.find({ city: "Bangalore", age: { $lte: 19 } });
{ "_id" : ObjectId("68c7b53642e9879892378991"), "name" : "Priya", "age" : 19,
"gender" : "female", "class" : "SE B", "CGPA" : 8.1, "city" : "Bangalore",
"course" : "IT", "graduated" : true }

> db.student.updateOne(
...   { name: "Sayali" },
...   { $set: { course: "MCA" } }
... );
{ "acknowledged" : true, "matchedCount" : 1, "modifiedCount" : 1 }

> db.student.deleteOne({ age: 18, gender: "male" });
{ "acknowledged" : true, "deletedCount" : 1 }

> db.student.updateMany(
...   { CGPA: { $gte: 6.5 } },
...   { $set: { graduated: true } }
... );
{ "acknowledged" : true, "matchedCount" : 8, "modifiedCount" : 3 }

> db.student.find({ CGPA: { $gt: 8.0 }, graduated: true });
{ "_id" : ObjectId("68c7b53642e9879892378991"), "name" : "Priya", "age" : 19,
"gender" : "female", "class" : "SE B", "CGPA" : 8.1, "city" : "Bangalore",
"course" : "IT", "graduated" : true }
{ "_id" : ObjectId("68c7b53642e9879892378995"), "name" : "Anjali", "age" : 20,
"gender" : "female", "class" : "TE A", "CGPA" : 9, "city" : "Bangalore",
"course" : "CS", "graduated" : true }
{ "_id" : ObjectId("68c7b53642e9879892378998"), "name" : "Suresh", "age" : 21,
"gender" : "male", "class" : "BE B", "CGPA" : 8.5, "city" : "Bangalore",
"course" : "Civil", "graduated" : true }

> db.student.deleteMany({ graduated: false });
{ "acknowledged" : true, "deletedCount" : 2 }

> db.student.find({
...   $nor: [
...     { course: "CS" },
...     { CGPA: { $gt: 3.5 } }
...   ]
... });

> db.student.deleteMany({ CGPA: { $lt: 3.0 } });

```

```
{ "acknowledged" : true, "deletedCount" : 0 }
```

```
> db.student.drop()  
true
```

```
> use studentDB  
switched to db studentDB
```

```
> db.dropDatabase()  
{ "dropped" : "studentDB", "ok" : 1 }
```